

Deep Learning and Swarm Intelligence for Boost PFC: Neural Network Voltage Control with PSO-Optimized Sliding Mode

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Abstract— In this paper, a boost power factor correction (PFC) is proposed with focus on output voltage and input current controllers. The output voltage of the boost converter needs to be immune against high load changes to prevent overshoot or undershoot on the load side. Moreover, the input current controller is to be robust against disturbances and uncertainties to have a satisfactory current in the source side. Therefore, a deep neural network voltage controller is proposed to guarantee fast dynamic response. Furthermore, a sliding mode controller (SMC) based on error and error integral is used to mitigate the chattering of the PFC's input current and ensure high power quality. A PSO-based optimization algorithm is also used to tune the SMC coefficients to enhance the performance of the boost PFC. Simulation results show satisfactory performance of both voltage and current controllers under steady state and load change circumstances.

Keywords— Boost Power Factor Correction, Deep Neural Network, Sliding Mode Controller, Optimization Algorithm

I. INTRODUCTION

Diode rectifiers are used to convert AC source voltage into DC form various range of applications. This rectification, however, creates considerable current harmonics due to pulse-shaped current and reactive power. Power factor Correction (PFC) converters are a sort of power quality improvement devices which tend to provide a DC voltage at the output while keeping the input current sinusoidal in-phase with source voltage, i.e. low harmonic and reactive power [1, 2]. PFCs are widely used in motor drive rectifiers, electrical vehicle chargers, and other industrial applications where DC voltage with high current is required [3]. From the point of view of control scheme of PFC, voltage control is essential to be accurate to reduce the voltage ripple in the output and fluctuations when sudden load changes occur. On the other hand, current control scheme mainly focuses on reducing current ripple to gain lower size of converter inductor, electromagnetic interface filters, and higher power quality [4].

Boost converter is a common structure which is used when a voltage higher than the input voltage is needed. Moreover, boost converter results in continuous input current which is another privilege compared to other structures [5]. Since boost converter is naturally a non-linear converter, non-linear controllers have been developed to control voltage and current of this converter, such as model predictive control [6], sliding mode control (SMC) [7], fuzzy control [8], and Neural Network (NN) [9] based controls. Among these controllers,

SMC has been used frequently due to the advantages it offers, i.e. simple implementation, robustness against disturbances and uncertainties, acceptable stability, etc. Various kinds of SMC have been studied in recent literature to improve its performance which led to terminal SMC, super-twisting SMC, adaptive SMC, fuzzy SMC, etc. These methods differ on the sliding surface and control law selection, chattering, and other factors [10]. To superior operation of SMC, its existing parameters must be optimized throughout system function. Various methods have developed optimization algorithms, e.g. Particle Swarm Optimization (PSO), Genetic Algorithm (GA), etc. to tune the controller gains [11]. In [12], PSO algorithm is used to obtain the optimum gains for SMC, especially for controlling the variable step of the MPPT algorithm in a multi-level PV inverter. In [13], a tuning approach with PSO is proposed for second order SMC which resulted in lower integral absolute error and integral square error. In this paper, PSO algorithm is utilized to tune the SMC coefficients to gain lower chattering and better dynamic response.

PI controllers are frequently used for voltage regulation in power electronic devices. However, they often demonstrate low accuracy of tracking and slow response, leading to low efficiency operation especially in critical condition. In [14], an approach has been facilitated to minimize these phenomena, applying a neural network-based STATCOM with high capability of adjusting bus voltage and power factor. Moreover, Artificial Neural Network (ANN) applications when comes with Model Predictive Control (MPC) in power converters can decrease computational burden and approximation of system model that MPC solely faces, so that making a way to easy implementation in practical subjects [15,16]. Although MPC strategies have been employed approximately in the most PFC structures, they also witness with lack of accuracy in higher-order system models [17]. This shortage can be ameliorated with the usage of NNs, which generally, Deep Neural Network-MPC (DNN-MPC) provides better dynamic response and tracking features than shallow layer network in nonlinearities [18].

In this paper, a boost PFC is proposed to provide DC output voltage for the load with high power quality. Motivated by the Advancement of NN in power electronics control, this article proposes a voltage controller based on neural network learning, showcasing high capability of tracking and faster dynamic response of this strategy compared to the

conventional PI controller. The current controller, on the other hand, utilizes a SMC controller to gain high robustness against uncertainties, low chattering, and satisfactory dynamic response.

The rest of this paper is as following: In section II, boost PFC structure is introduced. In section III, control scheme for output voltage and input current controller is proposed. Section IV provides simulation results.

II. SYSTEM CONFIGURATION

Fig.1 shows the structure of boost PFC converter. In Fig.1, V_s and I_s are the source voltage and current, respectively. L_s , L , C , and R are source reactance, boost inductor, boost capacitor, and load resistance. i_L is inductor current and V_o is the output voltage.

In this section, the proposed control approach for boost PFC is scrutinized. Boost PFCs have two control loops, i.e. voltage control loop-outer loop, and current control loop-inner loop. The outer loop regulates the output voltage of PFC converter and moreover, provides the reference current for the inner loop. The inner loop, on the other hand, tracks the input references current to have unity power factor and low reactive power. In the following, the output voltage and input current controllers are expressed. Fig.2 shows the overall block diagram of the proposed boost PFC converter control scheme.

A. Output Voltage control loop based on DNN

In conventional PFC circuits, a PI controller regulates the output voltage by processing the voltage error and generating a input reference current amplitude. The conventional PI controller is expressed as:

$$I_{ref}(k) = K_p e_v(k) + K_i \sum_{j=0}^k e_v(j) \quad (1)$$

where e_v is the output voltage error, K_p is the proportional gain, and K_i is the integral gain.

The proposed DNN controller replaces this PI controller entirely, taking the output voltage error as input and producing the reference current amplitude as output, thereby eliminating the need for manual gain tuning (try and error) and providing superior dynamic response under load variations. The DNN

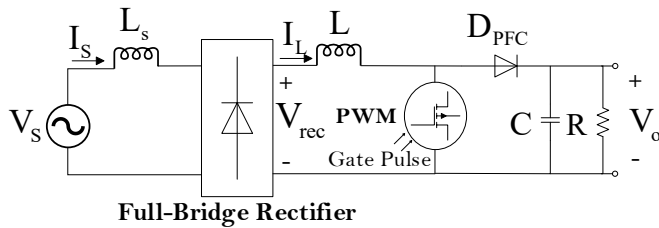


Fig. 1. Boost PFC Configuration

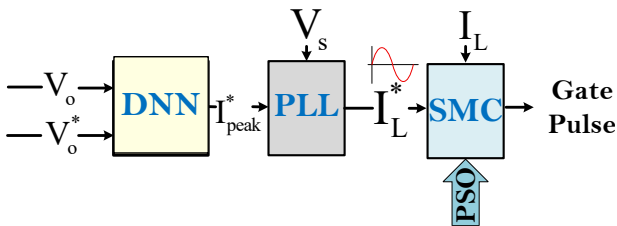


Fig. 2. Control Block Diagram of the Proposed Boost PFC

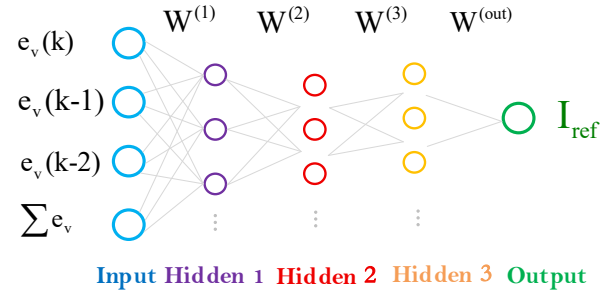


Fig. 3. DNN Architecture for Output Voltage Controller

architecture consists of an input layer, multiple hidden layer, and an output layer, as illustrated in Fig. 3. The network is designed to learn the optimal mapping from output voltage error to input reference current amplitude.

1) DNN Architecture

Input Layer: The DNN receives the output voltage error signal at time step k :

$$e_v(k) = V_{ref} - V_o(k) \quad (2)$$

To enhance dynamic performance, the input vector includes temporal information:

$$x(k) = [e_v(k), e_v(k-1), e_v(k-2), \sum_{j=k-N}^k e_v(j)]^T \quad (3)$$

where the first three terms provide derivative information through finite differences, and the fourth term represents the integral term.

The input features can be interpreted as:

- $e_v(k)$: Current output voltage error (proportional term)
- $e_v(k-1)$, $e_v(k-2)$: Past errors (derivative approximation)
- $\sum e_v(j)$: Accumulated error (integral term)

Hidden Layers: The proposed architecture employs three hidden layers. Each hidden layer applies a weighted sum followed by a non-linear activation function. For the l -th layer:

$$Z^{(l)} = W^{(l)} a^{(l-1)} + b^{(l)} \quad (4)$$

$$a^{(l)} = f(z^{(l)}) \quad (5)$$

where $W^{(l)} \in \mathbb{R}^{n_l \times n_{l-1}}$ is the weight matrix connecting layer $l-1$ to layer l , $b^{(l)} \in \mathbb{R}^{n_l}$ is the bias vector, $z^{(l)}$ is the pre-activation output, $a^{(l)}$ is the activation output, and $f(\cdot)$ is the activation function. Here, n_l denotes the number of neurons in layer l .

Activation Functions: In a neural network, without activation functions, all layers would only perform linear transformations. It doesn't matter how many layers are stack; the result would still be a linear function. In [19], the Rectified Linear Unit (ReLU) is introduced which employed in hidden layers due to its computational efficiency and ability to mitigate vanishing gradient problems. Activation functions introduce non-linearity, allowing the network to learn

complex patterns like the relationship between output voltage error and optimal input reference current in PFC controller.

Output Layer: The output layer produces the reference current amplitude(I_{peak}^*) using a linear activation function:

$$I_{peak}^*(k) = W^{(out)} a^{(3)} + b^{(out)} \quad (6)$$

where $W^{(out)}$ and $b^{(out)}$ are the output layer weights and bias. The output represents the input reference current amplitude that would have been produced by an optimal PI controller. The output is constrained to physically realizable values through saturation:

$$I_{peak}(k) = sat(I_{peak}^*(k), I_{min}, I_{max}) \quad (7)$$

where $I_{min} = 0$ and I_{max} corresponds to the maximum converter input current rating.

2) Training Methodology

In this intersection, the DNN is trained to minimize the output voltage error thanks to the applying Backpropagation Algorithm for loss function and the Adam (Adaptive Moment Estimation) [20] optimizer is employed for weight updates due to its adaptive learning rate capabilities and fast convergence, while maintaining fast transient response. Training datasets are generated through simulation of the PFC circuit under various operating conditions:

- Load steps: 25%, 50%, 75%, 100% rated power
- Source voltage variations: $\pm 15\%$ nominal
- Reference output voltage steps: $\pm 10\%$ nominal

Each scenario is simulated using a conventional PI controller to generate input-output pairs. The dataset comprises 50,000 samples with 80% allocated for training, 10% for validation, and 10% for testing.

B. Sliding Mode for Input Current Control

Sliding mode controller (SMC) has become a popular approach for non-linear systems, especially when there is uncertainties. The main concept of SMC is to design a sliding surface (S) and then find a proper control signal to make S to move towards 0. Therefore, assuming that $S \neq 0$, the control signal should propel S to zero and then, keep in the vicinity of it. The former phase, i.e. conducting S to 0 is called reaching phase and the latter is referred to as sliding mode. In order to

implement SMC, the state-space model of the boost converter is written as:

$$\begin{cases} \frac{di_L}{dt} = \frac{V_{rec}}{L} - \frac{1-d}{L} i_L \\ \frac{dV_o}{dt} = \frac{(1-d)i_L}{C} - \frac{V_o}{RC} \end{cases} \quad (8)$$

In order to simplify the process, $u = 1 - d$ is replaced. Thus, i_L dynamics can be expressed as:

$$\frac{di_L}{dt} = \frac{V_{rec}}{L} - \frac{u}{L} i_L \quad (9)$$

Choosing the sliding surface, S , is of a great importance in SMC design and can affect the performance of the whole current control system. The simplest sliding surface is usually $S = e$, where $e = i_L - i_L^*$. However, more complicated sliding surfaces are proposed to improve the performance and reduce chattering. Hence, S is considered as:

$$S = e + K_2 \int e dt \quad (10)$$

$$e = i_L - i_L^* \quad (11)$$

Where i_L^* is the input reference current generated by voltage controller. Now that the sliding surface is determined, the control signal is to be achieved. The control signal, u , consists of two separate terms, i.e. a term for reaching phase, u_{reach} , and a term for sliding phase, u_{eq} . u_{reach} is usually a sign function that ensures $S\dot{S} < 0$. Although, sign function can lead to discrete control signal and chattering. Thus, smoother functions such as tanh are often used instead:

$$u_{reach} = K_1 \tanh(S) \quad (12)$$

Where K_1 is the gain of reaching phase. On the other hand, u_{eq} is obtained such that S remains on zero. Therefore, the S is used to calculate u_{reach} :

$$\dot{S} = \dot{e} + K_2 e = 0 \quad (13)$$

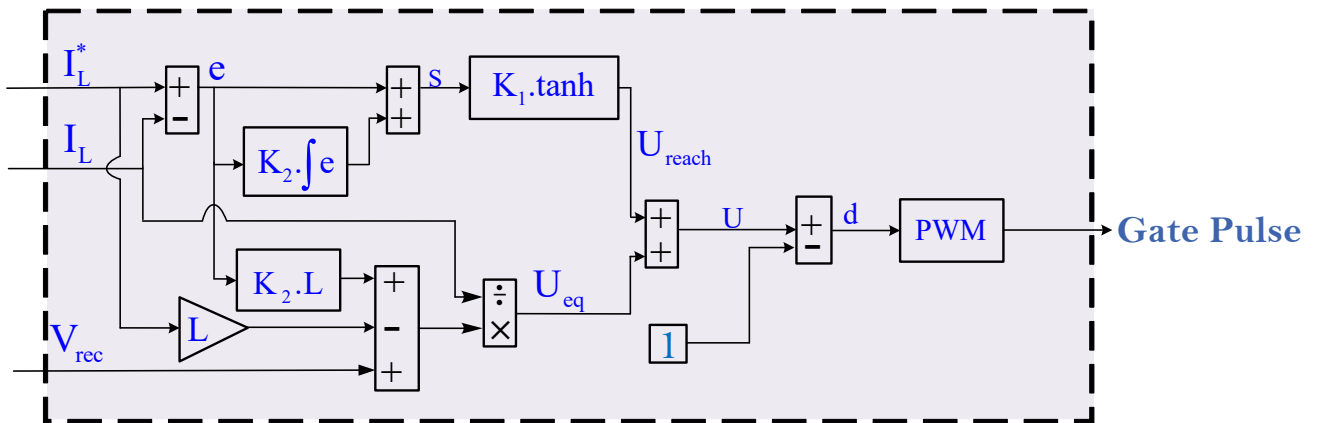


Fig. 4. Sliding Mode Controller for boost PFC

Where e is calculated from . Hence:

$$\dot{i}_L - \dot{i}_L^* + K_2(i_L - i_L^*) = 0 \quad (14)$$

where $\dot{i}_L$ is the derivative of i_L , and $\dot{i}_L^*$ is the derivative of the input reference current. Using (9) and (11), (14) can be re-written as:

$$\frac{V_{rec}}{L} - \frac{u_{eq}}{L} i_L - \dot{i}_L^* + K_2 e = 0 \quad (15)$$

$$u_{eq} = \frac{V_{rec}}{i_L} - \frac{L}{i_L} \dot{i}_L^* + \frac{K_2 L e}{i_L} \quad (16)$$

Furthermore, using (12) and (16), the final control signal is obtained as:

$$u = u_{reach} + u_{eq} \quad (17)$$

Remembering that the duty cycle, d , is to be applied to the power switch, using (17), d is calculated:

$$d = 1 - u \quad (18)$$

This duty cycle is then applied to a PWM block to generate gate pulses for the power switch. Fig.4 demonstrates the overall block diagram of the proposed SMC.

C. Tuning SMC Parameters (K_1, K_2) With PSO

The performance of SMC heavily depends on the proper selection of control gains K_1 and K_2 . PSO uses a swarm of particles, where each particle i represents a candidate solution $\mathbf{x}_i = [K_1, K_2]^T$ [21]. The algorithm updates particle positions based on:

$$V_i^{k+1} = w(k) \cdot V_i^k + c_1 r_1 (p_{best,i} - x_i^k) + c_2 r_2 (g_{best} - x_i^k) \quad (19)$$

Where v_i , $p_{best,i}$, g_{best} , r_1 , r_2 , c_1 , c_2 and $w(k)$ are particle velocity, personal best position, global best position, random numbers, acceleration coefficients and adaptive inertia weight, accordingly.

$$x_i^{k+1} = x_i^k + V_i^{k+1} \quad (20)$$

Each particle (K_1, K_2) is evaluated by simulating the PFC circuit under all test scenarios (section A.2) and must predominantly be chosen based on minimum grid current THD in each iteration. The primary goal is attaining grid current THD at lower value, therefore, multi-objective cost function is defined as follows:

$$J(k_1, k_2) = THD_i + 0.1(w_1 \cdot ITAE_i + w_2 \cdot Ripple_i + w_3 \cdot Chattering) \quad (21)$$

Where:

$$ITAE_i = \int_0^T t |i_L(t) - i_{ref}(t)| dt \quad (22)$$

$$Ripple_i = \sqrt{\frac{1}{T} \int_0^T (i_L(t) - \bar{i}_L)^2 dt} \quad (23)$$

$$Chattering = \frac{1}{T} \int_0^T \left| \frac{du}{dt} \right| dt \quad (24)$$

THD_i is the primary goal, w_1 , w_2 , and w_3 are the weighting factors, prioritizing tracking accuracy while limiting ripple and chattering. While pure THD minimization achieves marginally better harmonic performance, the proposed PSO approach balances power quality with

dynamic performance. By incorporating ITAE, ripple, and chattering into the cost function with appropriate weighting, the optimizer finds solutions to ameliorates grid current THD at low value, better dynamic response and less component stress. The PSO iteratively updates particles until convergence or maximum iterations.

III. SIMULATION RESULTS

In this section, the simulation results are provided for two various loads, i.e. 50Ω and 150Ω to verify the performance of the proposed control approach. The comparison is conducted in both steady state and dynamic responses. The system parameters are as expressed in Table.1.

A. Steady state performance

Fig.5 provides V_o regulated with PI controller with 150Ω load. As it can be seen, there is a fluctuation of 2% in the output voltage, which is the best performance of PI, i.e. least load at the output. Fig.6 and Fig.7, on the other hand, demonstrate V_o with DNN controller for 150Ω, and 50Ω loads, respectively. It can be seen that in this circumstance, the output voltage has only 1.25% and 2.5% fluctuations for 150Ω and 50Ω, respectively. This proves the superior performance of the proposed DNN algorithm, even under heavy load. Fig.8 shows source voltage and current. V_s is only plotted to check the in-phase current and therefore, is scaled. As it can be observed, the source current is absolutely in-phase with source voltage and moreover, it is sinusoidal with THD=11.6%. Fig.6, on the other hand indicates V_s and I_s for 50Ω load. It can be observed that increasing the load results in better input current tracking. THD in this condition is approximately 4.29%.

B. Dynamic Response

In this part, the dynamic responses of the output voltage and input current controller are provided and compared. For this purpose, the load is suddenly changed from 50Ω to 150Ω and then to 50Ω again, to investigate all possible states. The load change moment is highlighted with an orange ellipse in all figures. Fig. 10 shows the output voltage behavior with load change from 50Ω to 150Ω with PI controller. As it can be seen, there is a fluctuation of about 25% when the load is changed, which can seriously damage system devices. Fig.11

TABLE I. SYSTEM PARAMETERS

Parameter	Value
$V_{s, rms}$	220v
L_s	10μH
L	500μH
C	1000μF
R_{load}	50 Ω – 150 Ω
$V_{o, reference}$	400v
T_s	100μs
Optimized SMC coefficients for R=50 Ω	K1 = -2.1e-7 K2 = -10506
Optimized SMC coefficients for R=150 Ω	K1 = -2.29e-7 K2 = -10773

and Fig.12 illustrate the DNN performance when load changes from 50Ω to 150Ω and from 150Ω to 50Ω , respectively. It is obvious that the voltage controller tracks the reference output voltage almost with neither overshoot nor undershoot. Fig. 13 and Fig.14 indicate the input current controller performance under load change from 50Ω to 150Ω and vice versa, respectively. It can be seen that the proposed SMC tracks the input reference current properly with the least deviation from it.

C. Comparative Evaluation

Table II presents a comprehensive comparison between the proposed control strategy and existing PFC control approaches from recent literature. The proposed DNN-PSO framework demonstrates significant quantitative improvements across all performance metrics. Compared to conventional PI-SMC [7], fuzzy-based approaches [8], ANN-MPC methods [15], and single-objective PSO optimization [12], the proposed strategy achieves substantial reductions in steady-state voltage ripple, complete elimination of load transient overshoot, and improved current harmonic performance. The multi-objective PSO optimization ensures balanced performance between power quality and dynamic response, while the DNN voltage controller eliminates the trial-and-error tuning process required in conventional approaches.

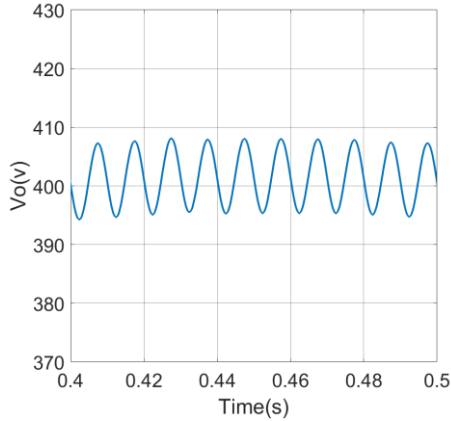


Fig. 5. V_o with PI controller with 150Ω load

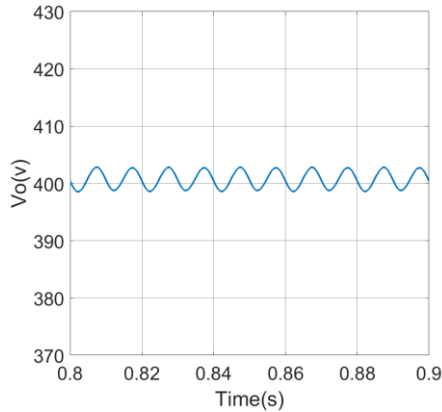


Fig. 6. V_o with DNN with 150Ω load

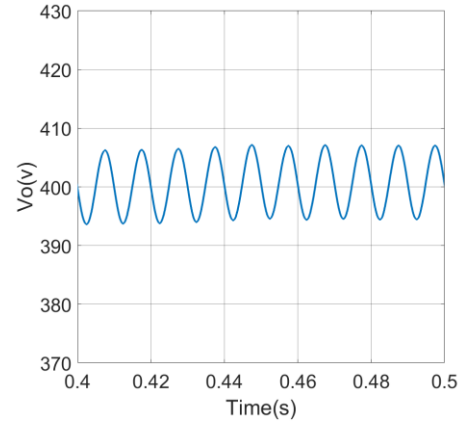


Fig. 7. V_o with DNN with 50Ω load

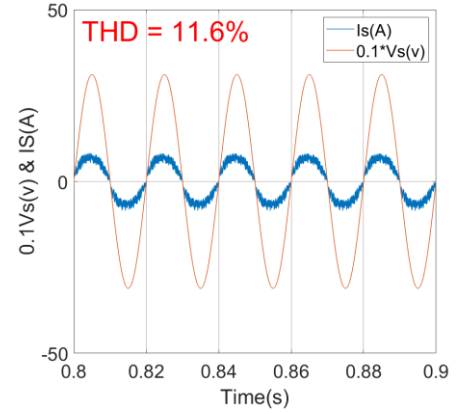


Fig. 8. V_s and I_s with 150Ω load

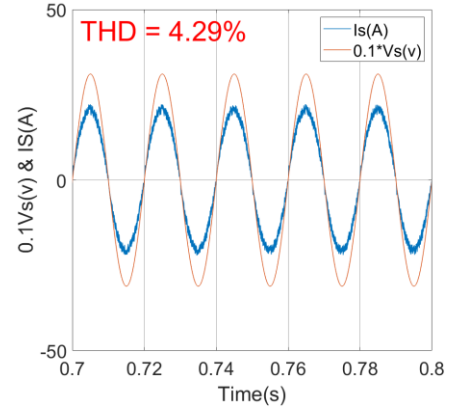


Fig. 9. V_s and I_s with 50Ω load

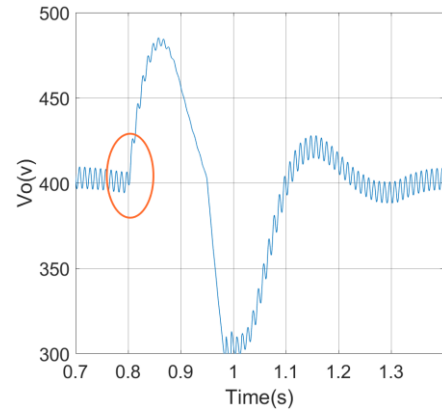


Fig. 10. V_o variation with load from 50Ω to 150Ω with PI

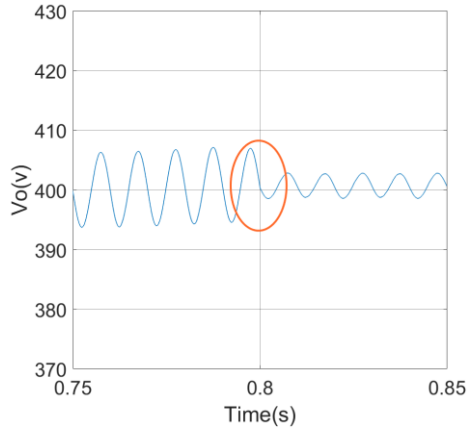


Fig. 11. V_o variation with load from 50Ω to 150Ω with DNN

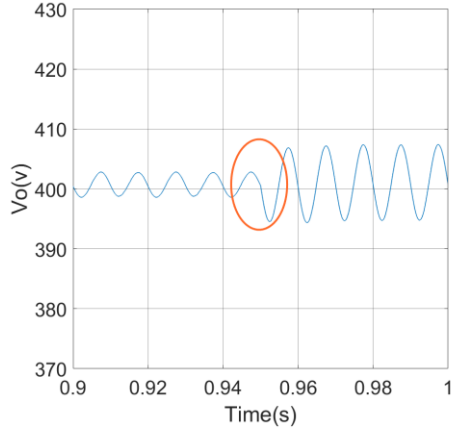


Fig. 12. V_o variation with load from 150Ω to 50Ω with DNN

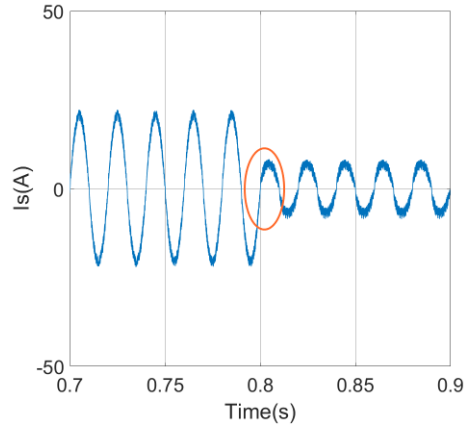


Fig. 13. Source current tracking with load change from 50Ω to 150Ω

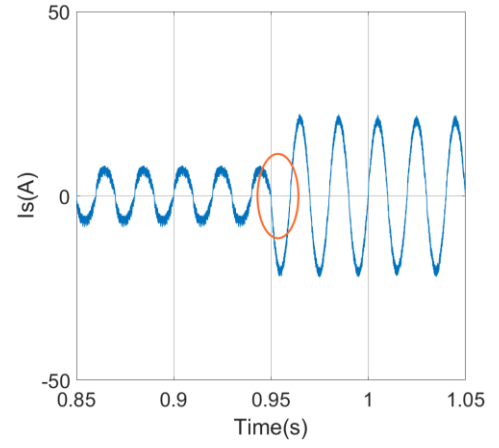


Fig. 14. Source current tracking with load change from 150Ω to 50Ω

IV. CONCLUSION

This paper presents a hybrid intelligent control strategy for boost PFC converters combining deep neural network output voltage regulation with PSO-optimized sliding mode current control. The DNN controller with multi-layer architecture and ReLU activation replaces conventional PI control, eliminating manual tuning while achieving superior output voltage regulation and transient response under load variations. A sliding mode controller ensures robust input current tracking with minimal chattering. PSO optimization employs a multi-objective cost function prioritizing THD minimization while balancing tracking accuracy, ripple, and chattering, yielding optimal SMC gains that maintain sinusoidal input current with low harmonic distortion across all operating conditions. The proposed framework demonstrates three key advantages: elimination of trial-and-error controller tuning through intelligent optimization, superior dynamic performance during load transients compared to PI, Fuzzy, ANN-based MPC control, and excellent power quality compliance with international standards. This synergistic integration of deep learning and swarm intelligence provides a robust, adaptive solution for modern PFC applications in electric vehicle chargers, motor drives, and industrial power supplies. Future work includes experimental hardware validation, online adaptive training, and extension to three-phase high-power systems.

TABLE II. PERFORMANCE COMPARISON WITH STATE-OF-THE-ART CONTROL STRATEGIES

Reference	Voltage/Current Control Strategy	Voltage Overshoot /Undershoot (%)	THD Under Light Load(%)	Settling Time (%)	Optimization
[7]	PI / Sliding mode	5% Undershoot	< 3%	~100ms	Manual tuning
[8]	Fuzzy PID /Conventional PID	15% Overshoot	< 12%	Slow response	Trial-and-error
[15]	MPCANN-based MPC/ ANN-based MPC	12%	5.5%	Fast Response	Offline Training
[12]	Conventional PI/PSO-adaptive SMC	~18% Overshoot	<7.2%	not specified	PSO(single objective)
Proposed	DNN/PSO-SMC	~0%	4.29%	<10 ms	PSO(multi-objective, THD-focused)

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